15/01/25 MIS

USN | | | | | | | | | | | [24BSCSMA11]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2025

COMPUTATIONAL MATHEMATICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **(i) Missing data may be suitably assumed**

(ii) Answer all Questions

			Blooms Level	CO	Marks
		SECTION – A			**5 X 2=10**
1.	**a.**	Let $A = \{2,3,4\}$ and $B = \{4,5\}$. Find **i.** Find the number of relations from A to B **ii.** Find the number of relations from B to A.	L1	CO1	**02**
	b.	Write negation of the following statement. **i.** $p = 2 + 5 < 1$. **ii.** It will snow tomorrow or it will rain tomorrow.	L2	CO1	**02**
	c.	Find the first order partial derivatives of $u = e^{2x}\, siny$.	L2	CO1	**02**
	d.	Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 4 \\ 4 & 4 & 7 \end{bmatrix}$.	L1	CO1	**02**
	e.	Find the general solution of $(D^2 - 1)y = 0$.	L2	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Explain different types of functions with suitable examples.	L2	CO1	**05**
		OR			
	b.	Consider $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x - 1}$ defined on $f, g: R \to R$. Show that $(g\ o\ f)(1) = 1, (g\ o\ f)(2) = 2$, $(g\ o\ f)\ (3) = 3$, and $(g\ o\ f)\ (x) = x$.	L3	CO1	**05**
3.	**a.**	Write the inverse, converse and contrapositive of the proposition "If you do your homework, you will not be punished".	L3	CO3	**05**
		OR			
	b.	Prove the following logical equivalence without using Truth table. $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \Leftrightarrow (p \vee q \vee r)$.	L3	CO3	**05**
4.	**a.**	If $z = e^{ax+by}\, f(ax - by)$ then show that $b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz$.	L3	CO6	**05**
		OR			
	b.	Find the directional derivative of $f = xy + xz + yx$ at (1, 2, 0) along $\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.	L3	CO6	**05**

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5.	**a.**	Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$ by converting it into row Echelon form.	L3	CO4	**05**
		OR			
	b.	Solve the following system of equations by Gauss elimination method $2x + y + 4z = 12;\ 4x + 11y - z = 33;\ \ 8x - 3y + 2z = 20.$	L3	CO4	**05**
6.	**a.**	Solve the differential equation $(D^2 + 5D + 6)y = e^{3x}$.	L3	CO5	**05**
		OR			
	b.	Find particular integral (y_p) corresponding to the differential equation $(D^2 + 4)y = 4\sin 2x + \cos 5x$.	L3	CO5	**05**
7.	**a.**	Examine whether the function $f(x, y) = x^3 + y^3 - 3xy + 1$ is minimum at the point (1,1) or not?	L3	CO6	**05**
		OR			
	b.	Find divergence and curl of $\vec{v} = (xyz)i + (3x^2y)j + (xz^2 - y^2z)k$ at (2, -1, 1).	L3	CO6	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Define Stirling functions. If there are six programmers, who can assist eight executives? How many ways can the executives be assisted so that each programmer assists at least one executive?	L3	CO1	**10**
		OR			
	b.	Let $f: R \to R$ be defined by $$f(x) = \begin{cases} 3x - 5 & for\ x > 0 \\ -3x + 1 & for\ x \leq 0 \end{cases}$$ **i.** Determine $f(0)$, $f(-1)$, $f\left(\frac{5}{3}\right)$, $f(-\frac{5}{3})$. **ii.** Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(3)$, $f^{-1}(-3)$, $f^{-1}(-6)$	L3	CO2	**10**
9.	**a.**	Write the Rules of Modus Tollens and Modus Ponens. Test the validity of the following arguments If Ravi goes out with friends, he will not study. If Ravi does not study, his father becomes angry His father is not angry ∴ Ravi has not gone out with friends	L3	CO3	**10**
		OR			

	b.	Define quantifiers and provide examples for the universal quantifier and the existential quantifier. Determine the truth value of each of the following quantified statements with one example each, the universe being the set of all non-zero integers. **i.** $\exists\, x, \exists\, y, [xy = 1]$ **ii.** $\exists\, x, \forall\, y, [xy = 1]$ **iii.** $\forall\, x, \exists\, y, [xy = 1]$ **iv.** $\exists\, x, \exists\, y, [(2x + y = 5)\Lambda(x - 3y = -8)]$ **v.** $\exists\, x, \exists\, y, [(3x - y = 17)\Lambda(2x + 4y = 3)]$	L3	CO3	**10**
10.	**a.**	Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$, where $u = log(x^3 + y^3 + z^3 - 3xyz)$ and hence show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.	L3	CO6	**10**
		OR			
	b.	Examine the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$.	L3	CO6	**10**
11.	**a.**	Find all the eigenvalues and the corresponding eigenvectors of the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ and hence find A^4.	L3	CO4	**10**
		OR			
	b.	Investigate the values of λ and μ such that the system of equations $x + y + z = 6;\ x + 2y + 3z = 10;\ x + 2y + \lambda z = \mu$, may have i] Unique solution, ii] Infinite Solution, iii] No solution.	L3	CO2	**10**
12.	**a.**	Solve $X' = \begin{bmatrix} -2 & -1 & 8 \\ 0 & -3 & 8 \\ 0 & -4 & 9 \end{bmatrix} X$ by diagonalization.	L3	CO5	**10**
		OR			
	b.	Solve the following differential equation $(D^2 - D)\, y = x^2 - 2x - 32.$	L3	CO4	**10**
13.	**a.**	Solve the following Legendre's differential equation $(1 + x)^2 \frac{d^2y}{dx^2} + (1 + x) \frac{dy}{dx} + y = 2 \sin[\log(1 + x)].$	L3	CO6	**10**
		OR			
	b.	Write python code for negation, conjunction and disjunction of two propositions p and q.	L3	CO5	**10**